PEPII

RF SYSTEM

RF STATION TURN-ON PROCEDURE

PS-340-330-59-R0

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RF Station Turn-On Procedure

Pre-condition: Klystron Filament & Solenoid "ON"
Filament Volt./Curr. "GREEN"
Filament Time Left 0 sec "GREEN"
(Klystron filament and solenoid can be turned ON from the EPIC control system *KLYSTRON* panel. Filament Time Left timer runs for 30 min., then interlocks can be cleared).

All boxes on main *RF STATION* panel "GREEN" except HVPS box if HVPS contactor is OFF.

Push ☛ **RESET**

Contactor ☛ **CLOSE** on main *RF STATION* panel or *HVPS* panel.

Note: If the Contactor cannot be closed several "manual reset" interlocks as listed on the HVPS panel may be faulted. These need to be reset using the RESET button on the HVPS SMART TOUCH panel in rack 2 of the electronic racks of each station.

Note: All following control buttons can be found on the *RF STATION* panel.

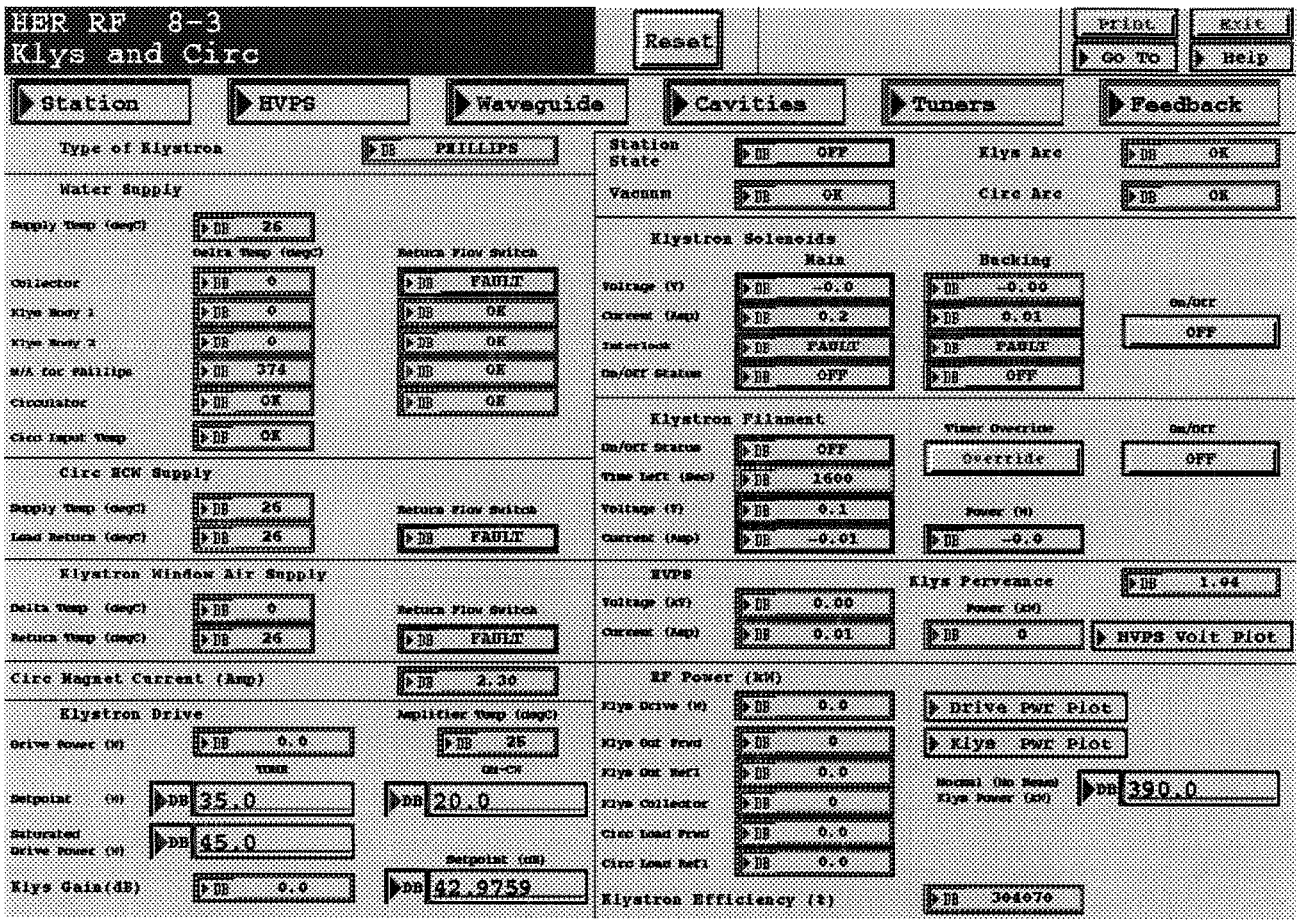
Beam operation:
Use ☛ **ON_CW** mode. HVPS Loop OFF/PROC/ON buttons in ☛ **ON**. The station will automatically run up the HVPS voltage to the set **Stn Gap Volt (kV)** and then regulate it there. Also the Ripple Loop, Direct Loop and Comb Loop will be initiated. (see: *FEEDBACK* panel for loop status).

Set **"Auto Reset Tries Left"** to a max. of 25 resets. This automatically brings the station back on after a trip except when the HVPS contactor trips.

Station parked:
To temporarily park station cavities ± 340 kHz from resonance go to ☛ OFF mode and then ☛ **PARK** mode with the HVPS contactor closed (all interlocks must be green to clear the beam abort and go to the PARK mode).

To long-term park the cavities a RF station can be changed into the OFF-LINE mode by pushing the ☛ **OFF-LINE** button on the *RF STATION* panel or by turning a **STATION LOCK-OUT key** at the **RF LOCAL DISPLAY PANEL** in rack 3 of the electronic racks of each station. Now the station can be put in the PARK mode with the HVPS and klystron turned off and the beam abort will only fire when essential interlocks like: waveguide pressure, cavity vacuum, cavity or load cooling flow interlocks trip. Note: If the station does not stay in the PARK mode one of the essential interlocks is not made up and the beam abort is not reset.

Fig. 1 KLYSTRON panel:



Note: The solenoid and the filament ON/OFF buttons are now combined into a single ON/OFF button.

Fig. 2 RF STATION panel:

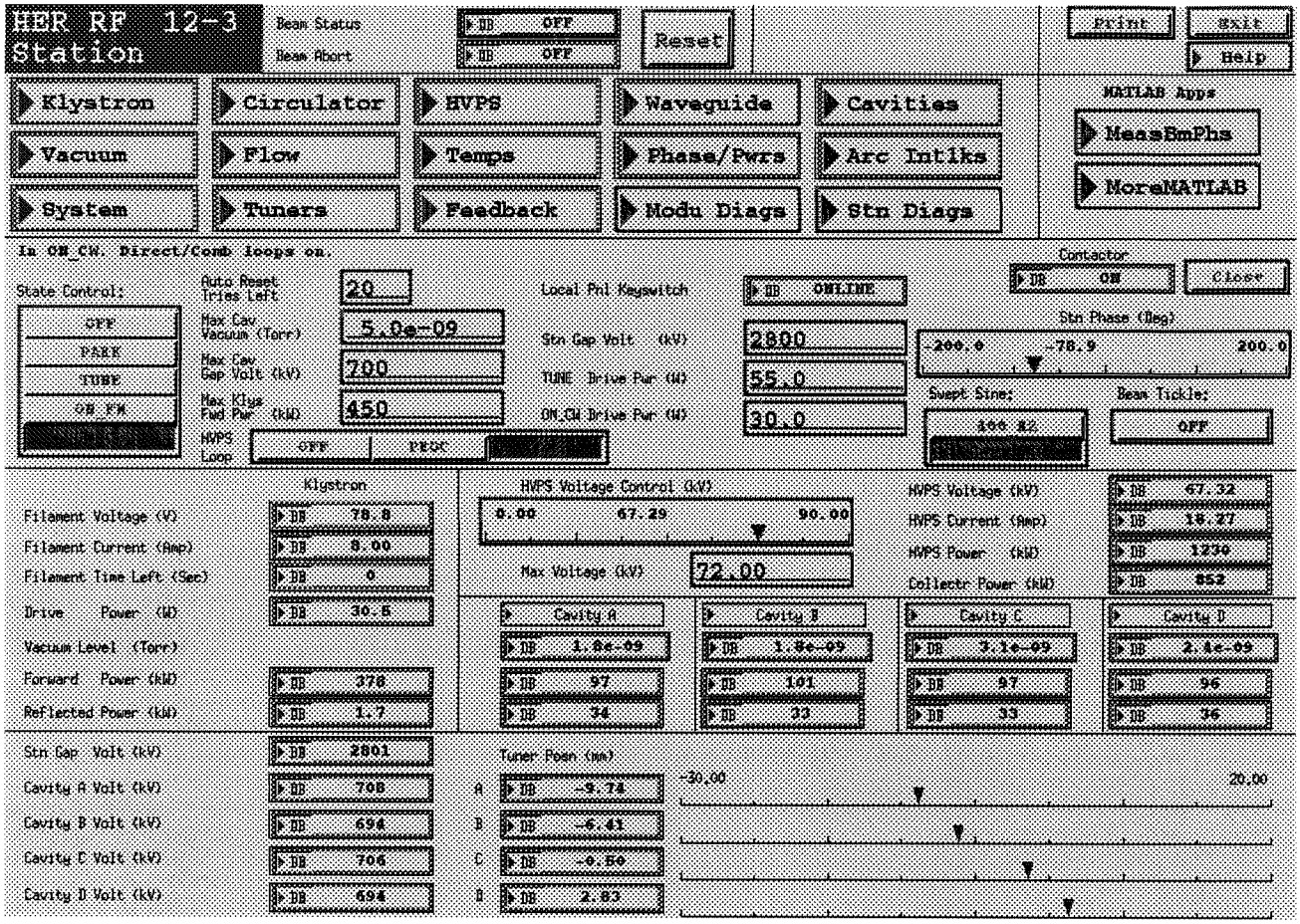


Fig. 3 Klystron HVPS panel:

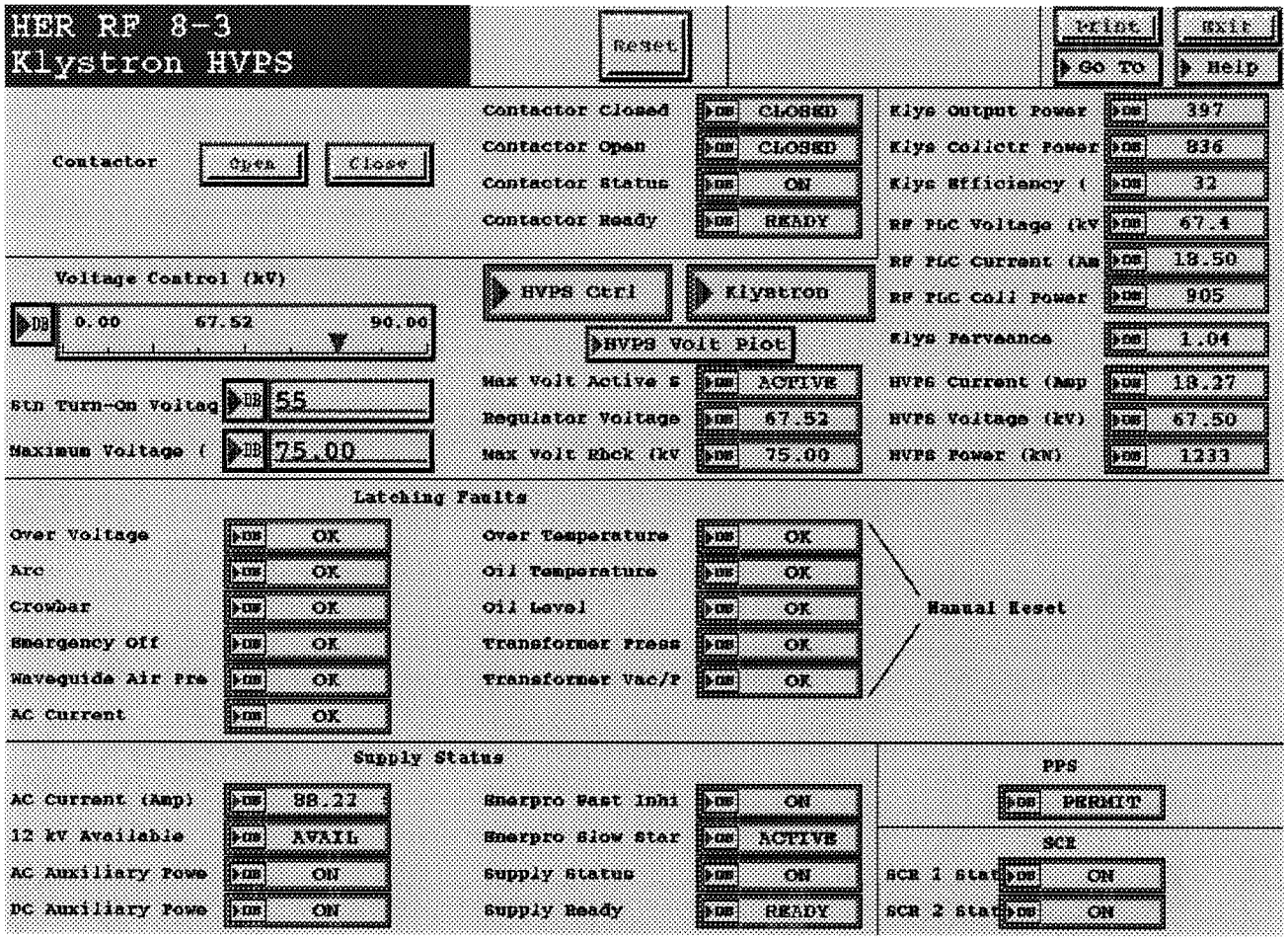
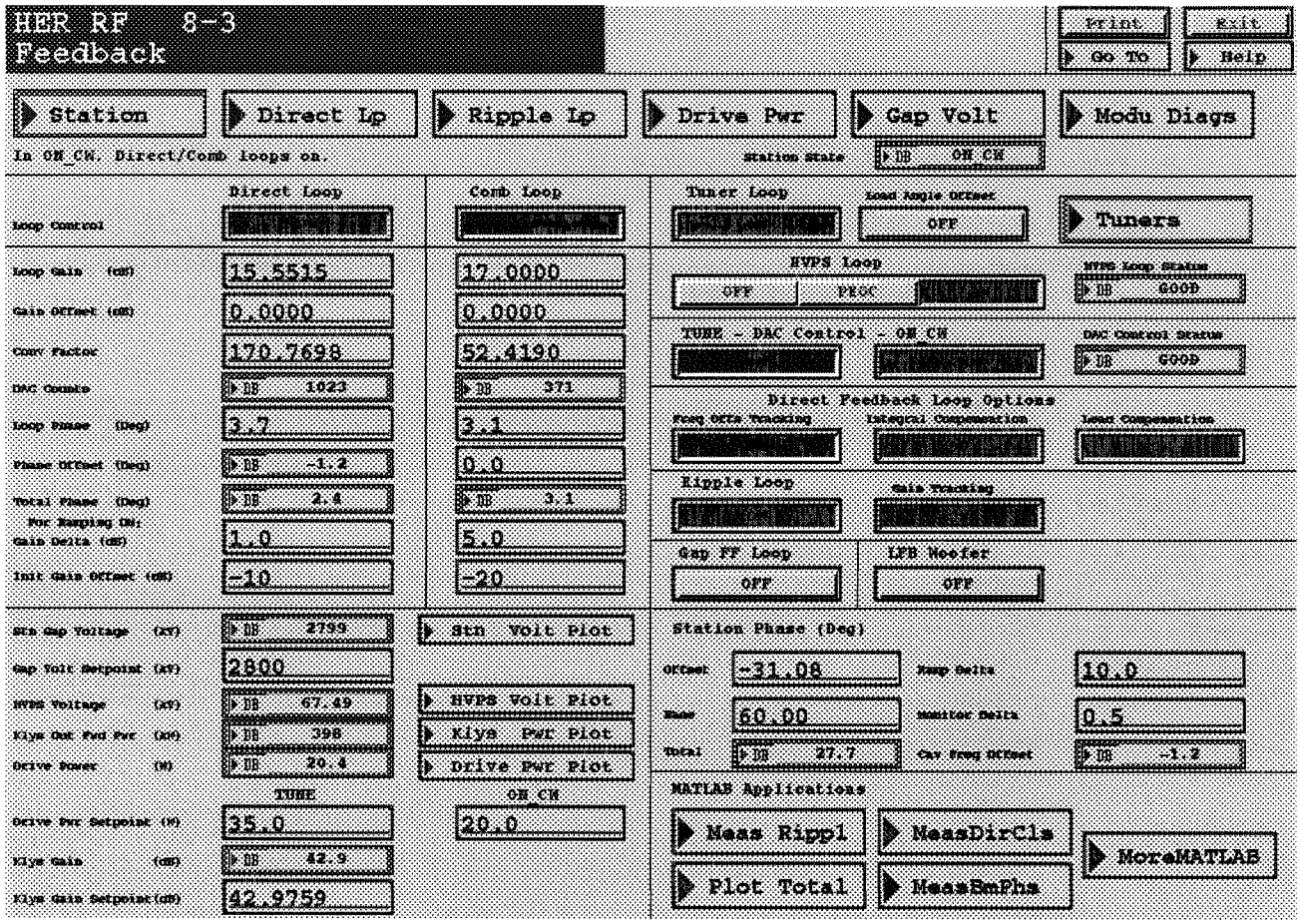


Fig. 4 Feedback panel:



OFF mode: Turns RF off with the HVPS contactor closed and fires beam abort, cavity tuners are left in their previous position.

Processing cavities:
After an intentional or accidental vent or long shutdown cavities are processed first in ON_FM mode at 1000 Hz and then in ON_CW mode using the HVPS Loop OFF/PROC/ON buttons in ☛ **PROC**. In the PROC mode the station will automatically step up the HVPS voltage until one of the limiting parameters is reached and steps the voltage down when the limit is passed.
The following table gives the nominal limit parameters:

Parameter	HER FM	HER CW	LER FM	LER CW
Max Cav Vacuum (Torr)	1E-8 Torr			
Max Cav Gap Volt (kV)	800 kV	750 kV	900 kV	850 kV
Max Klys Fwd Pwr (kW)	540 kW	450 kW	330 kW	290 kW